

SAMAD ZAIDI

AI / ML Engineer • Computer Vision • NLP • Production ML Pipelines

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PROFESSIONAL SUMMARY

Results-driven AI/ML Engineer with 2+ years delivering production-grade computer vision, NLP, and ML pipeline solutions for the Indian Army, state government bodies, industrial manufacturers, and global enterprise SaaS clients. Expertise spans YOLO object detection, CNN architectures, RAG pipelines, LLM integration (Claude API, OpenAI), FastAPI microservices, and full-stack ML deployment — with a proven track record of converting research prototypes into high-impact real-world systems with measurable outcomes.

PROFESSIONAL EXPERIENCE

AI Engineer

Eminence Technology, Mohali, India — Feb 2026 – Present

- German Healthcare Document Processing:** End-to-end FastAPI + Claude API pipeline converting raw/scanned German medical records into structured patient registration and insurance claim forms via dual-schema extraction; queue-based concurrent processing, AWS S3 integration, and automated email notifications eliminate manual data entry with high accuracy on German clinical terminology.
- BidHelper.ai — AI Bidding & Collaboration Platform:** Built the complete AI layer of an internal SaaS product — RAG pipeline (Pinecone + MongoDB), CSV-upload training portal with SEO/Non-SEO classification, intelligent URL fetcher for past-work context injection, humanised cover-letter generation with confidentiality guardrails, Improver Tab for iterative refinement, and a Trello-replica board with a context-aware bot per card that ingests JDs, PDFs, CSVs, client chats, and Loom links for instant non-technical team querying.

AI/ML Developer

Beyond Root Technologies, Mohali, India — Jun 2025 – Feb 2026

- PCB Defect Detection:** Real-time YOLO + SIFT homography QC for electric meter manufacturing; saved INR 2,500/unit with PostgreSQL-backed workflow.
- Automated Seal Verification:** CV pipeline with hardware jig; 100% placement accuracy, 40% rework reduction.
- SQL NL Chatbot:** Ollama Mistral interface for natural language to optimised SQL. | **Theft Detection:** OpenCV CCTV monitoring with React real-time alert dashboard.

AI/ML Developer

Stakza, Dubai, UAE (Remote) — Jun 2023 – May 2025

- Automated ML pipelines with cross-validation & hyperparameter optimisation; 15% accuracy improvement with production drift detection.
- ETL workflows on multi-terabyte datasets; automated reporting dashboards (Matplotlib, Seaborn) cut client info-gathering time by 60%.

AI & ML Engineer (Internship)

DDU Universe Pvt. Ltd., Chandigarh, India — Nov 2024 – Jan 2025

- Human-Elephant Conflict Detection (Indian Army):** CNN + Raspberry Pi real-time monitoring; 30% speed gain, 20% false-positive reduction in field deployment.
- All-Weather Pothole Detection (Punjab Govt):** CNN on robotic platform; ONNX/TensorRT 25% inference boost; GIS mapping for road maintenance planning.

KEY PROJECTS

BidHelper.ai — AI Bidding & Collaboration (2026)

Tech: RAG, Pinecone, MongoDB, FastAPI, LLM

- RAG + Pinecone/MongoDB; CSV training portal; SEO/Non-SEO classification; URL fetcher; humanised letter generation with guardrails; Improver Tab; Trello board bot ingesting JDs, PDFs, CSVs, Loom links for instant team querying.

Elephant Detection — Indian Army (2024)

Tech: YOLO, CNN, OpenCV, Raspberry Pi

- Multi-camera real-time tracking with geometric transformations; 20% false-positive reduction in active military zones.

German Healthcare Document Processing (2026)

Tech: FastAPI, Claude API, AWS S3

- Dual-schema extraction pipeline; async queue; S3 storage; automated admin notifications; POST API for DB submission.

Pothole Detection — Punjab Govt (2024)

Tech: YOLO, ONNX, TensorRT, GIS

- All-weather drone imagery model; 25% inference boost; GIS mapping for maintenance planning.

Scrape-n-Ask AI Chatbot (2024)

Tech: OpenAI API, Hugging Face

- Context-aware Q&A; and structured data extraction from diverse web sources.

TECHNICAL SKILLS

Languages & Frameworks: Python, SQL, TensorFlow, PyTorch, Scikit-learn, Keras, OpenCV, FastAPI, Django, Flask

Computer Vision: YOLO (v5/v8/v11), CNNs, Image Processing, SIFT, Homography, Object Detection & Tracking

Generative AI & NLP: LLMs, Claude API, OpenAI API, Ollama, Mistral, Hugging Face, Prompt Engineering, RAG, Pinecone

Data Science: Pandas, NumPy, Matplotlib, Seaborn, Feature Engineering, A/B Testing, Drift Detection

Deployment & Tools: ONNX, TensorRT, REST APIs, n8n, Docker, Azure, AWS S3, PostgreSQL, MongoDB, Git

Hardware & Edge: Raspberry Pi, IoT Devices, Robotic Platforms, Industrial Automation Systems

NOTABLE IMPACT METRICS

- INR 2,500 saved per unit — PCB defect elimination
- 100% seal placement accuracy via automated CV verification
- 40% rework time reduction via vision-based quality control
- 30% speed boost in real-time elephant conflict detection
- 25% inference boost via ONNX / TensorRT optimisation
- 60% drop in client info-gathering time via ETL dashboards
- 20% false-positive reduction in Indian Army field deployment
- 85%+ accuracy on RAG-based enterprise knowledge chatbot

EDUCATION & ACHIEVEMENTS

B.E. Computer Science — Information Security

Chandigarh University | CGPA: 7.80 / 10.0 | May 2025

DCPD Eminent Speaker Award — Chandigarh University

Recognised for outstanding technical communication in AI/ML.

Languages: English (Professional) • Hindi (Native)

Domains: Indian Army • Punjab Govt • Industrial Manufacturing • German Healthcare • Insurance • Enterprise SaaS • Open Source